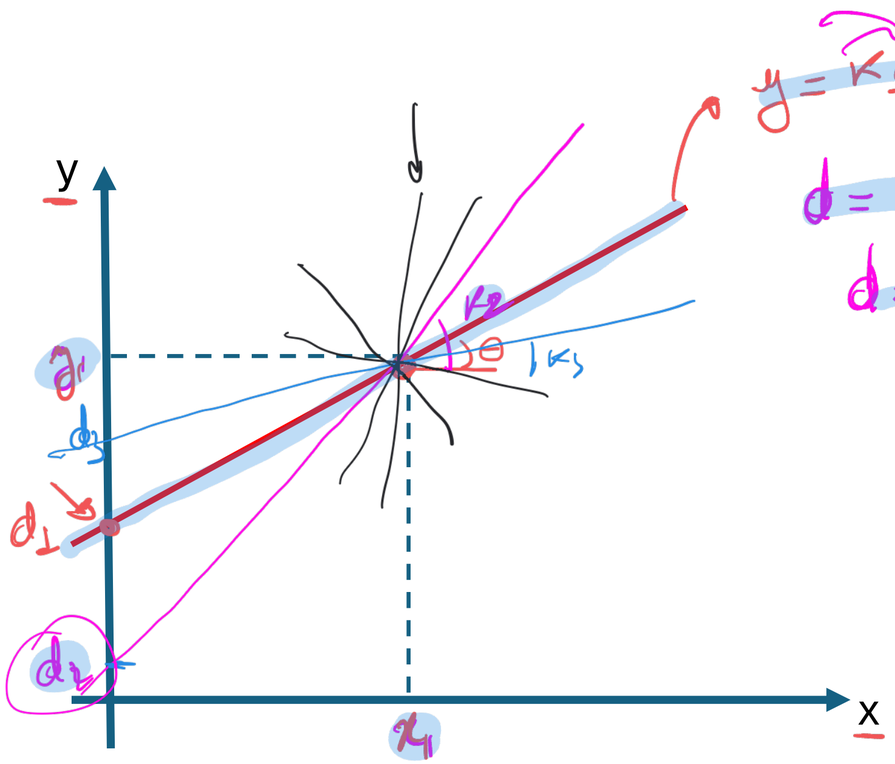
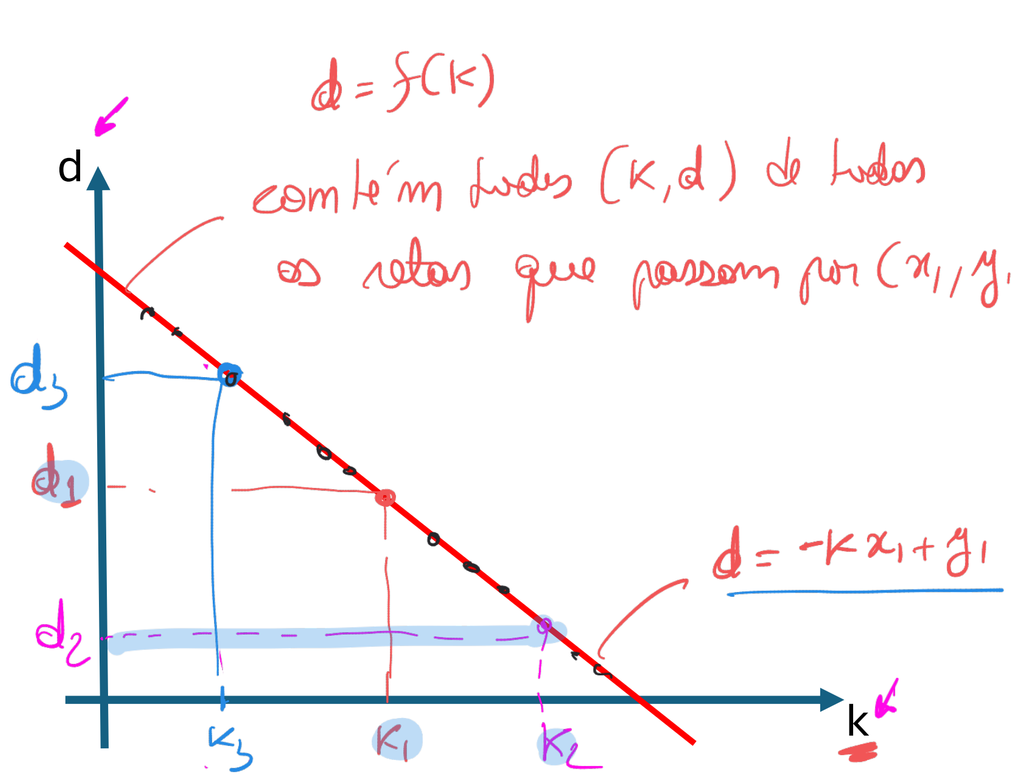
Image enhancement in the space domain

Hough Transform

2 (h 30 min



xy space domain



Hough Transform Parameters Space

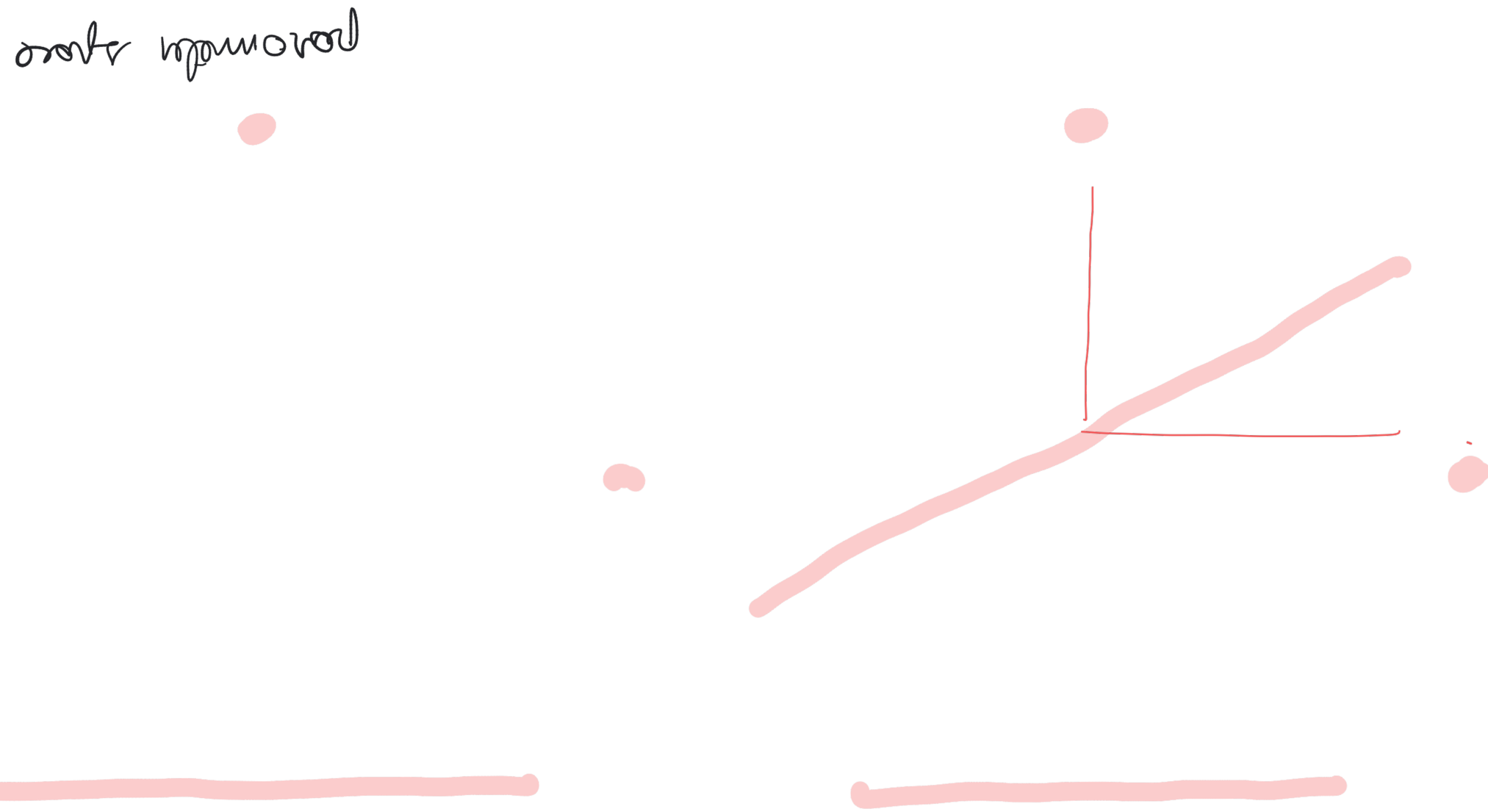
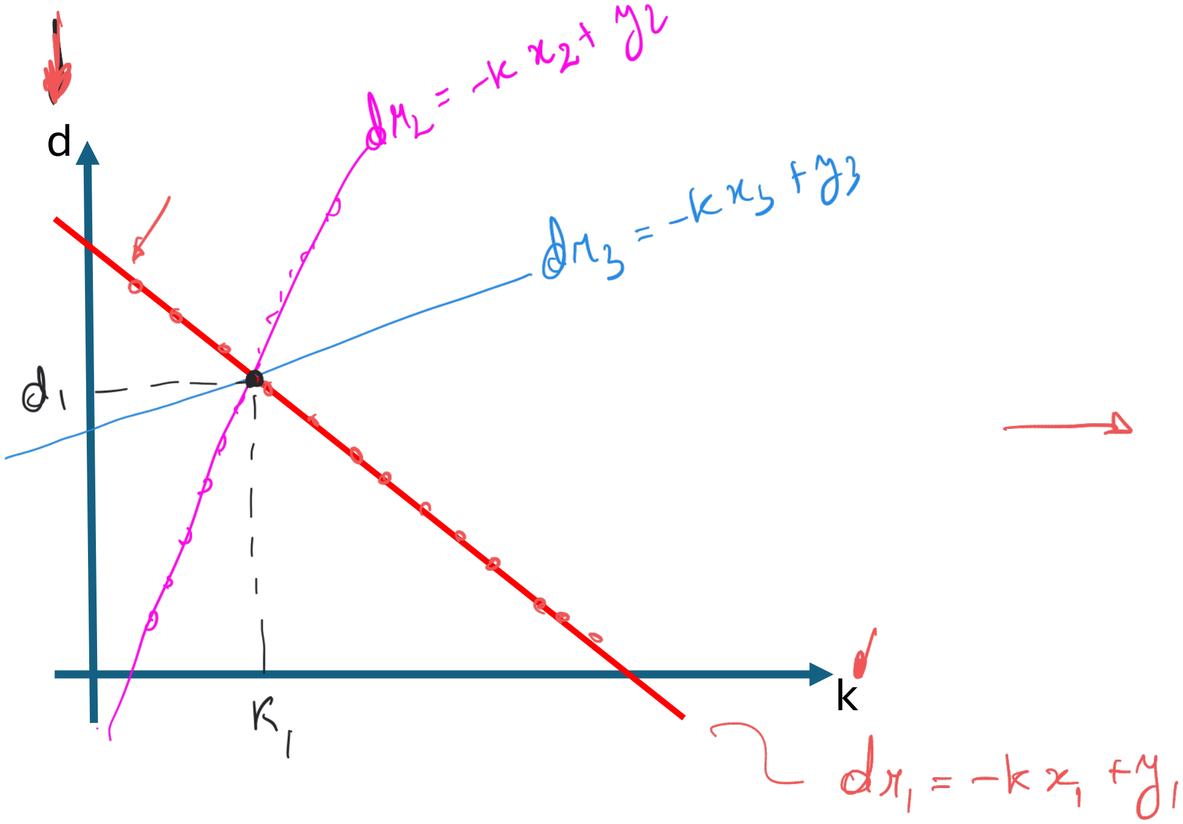
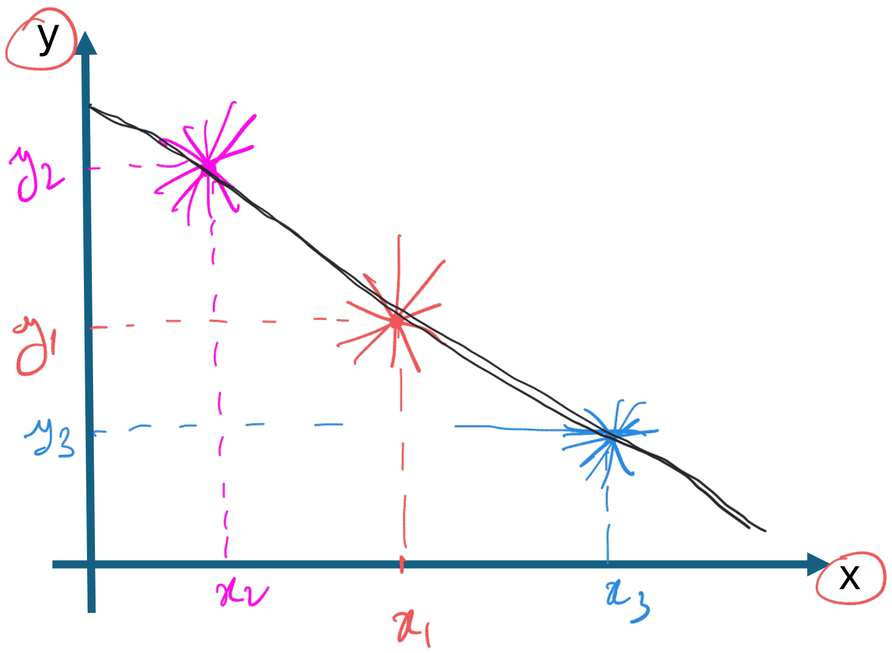


Image enhancement in the space domain

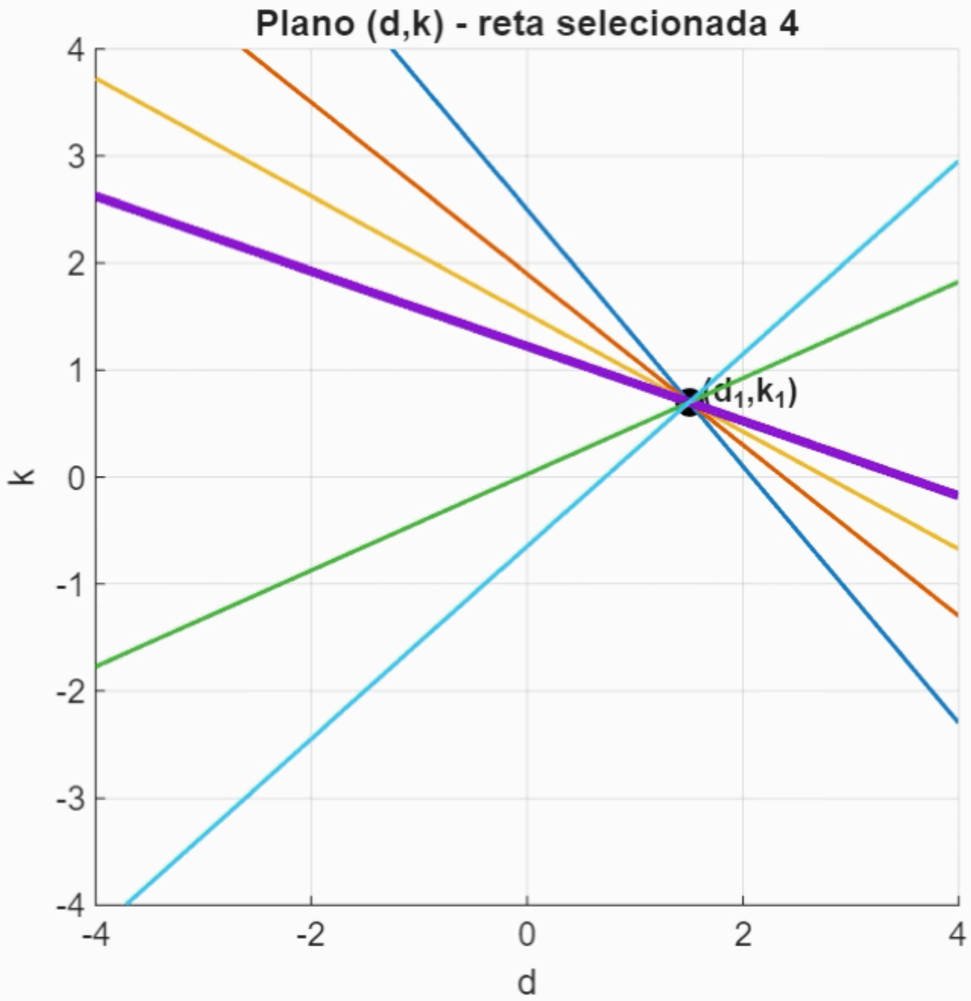
Hough Transform

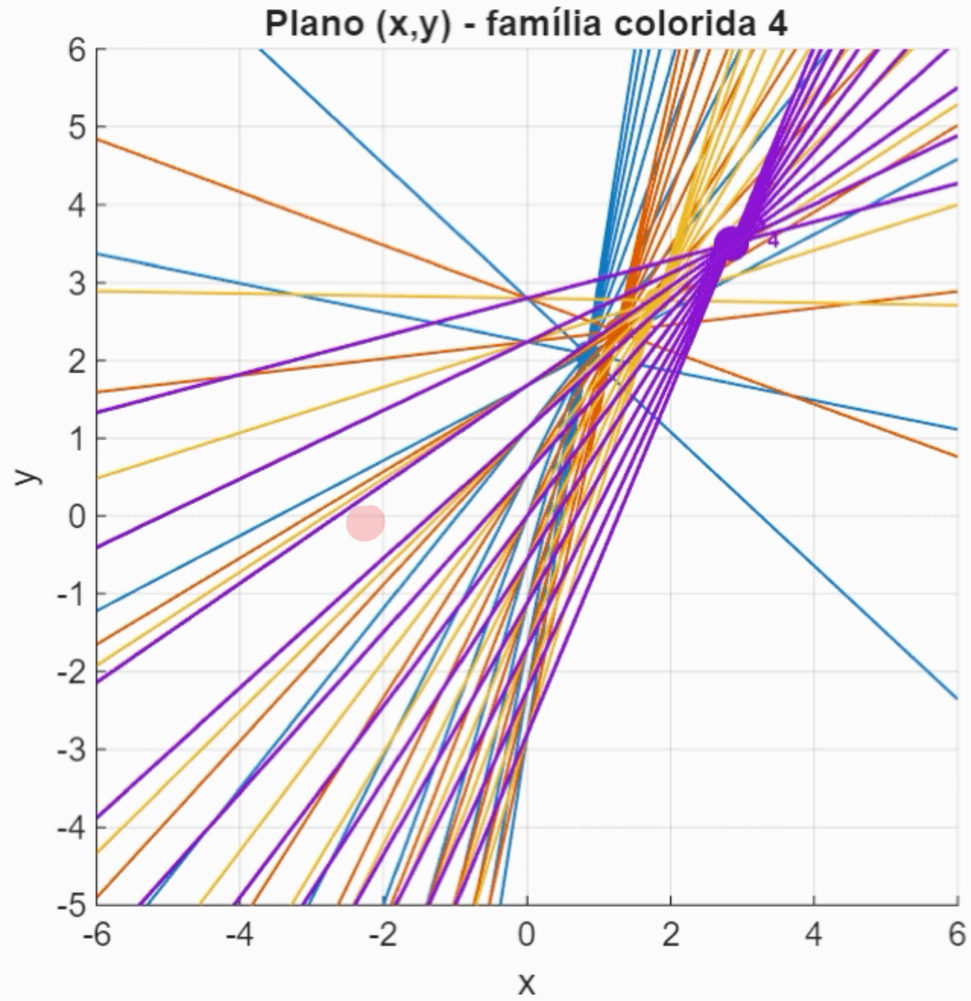


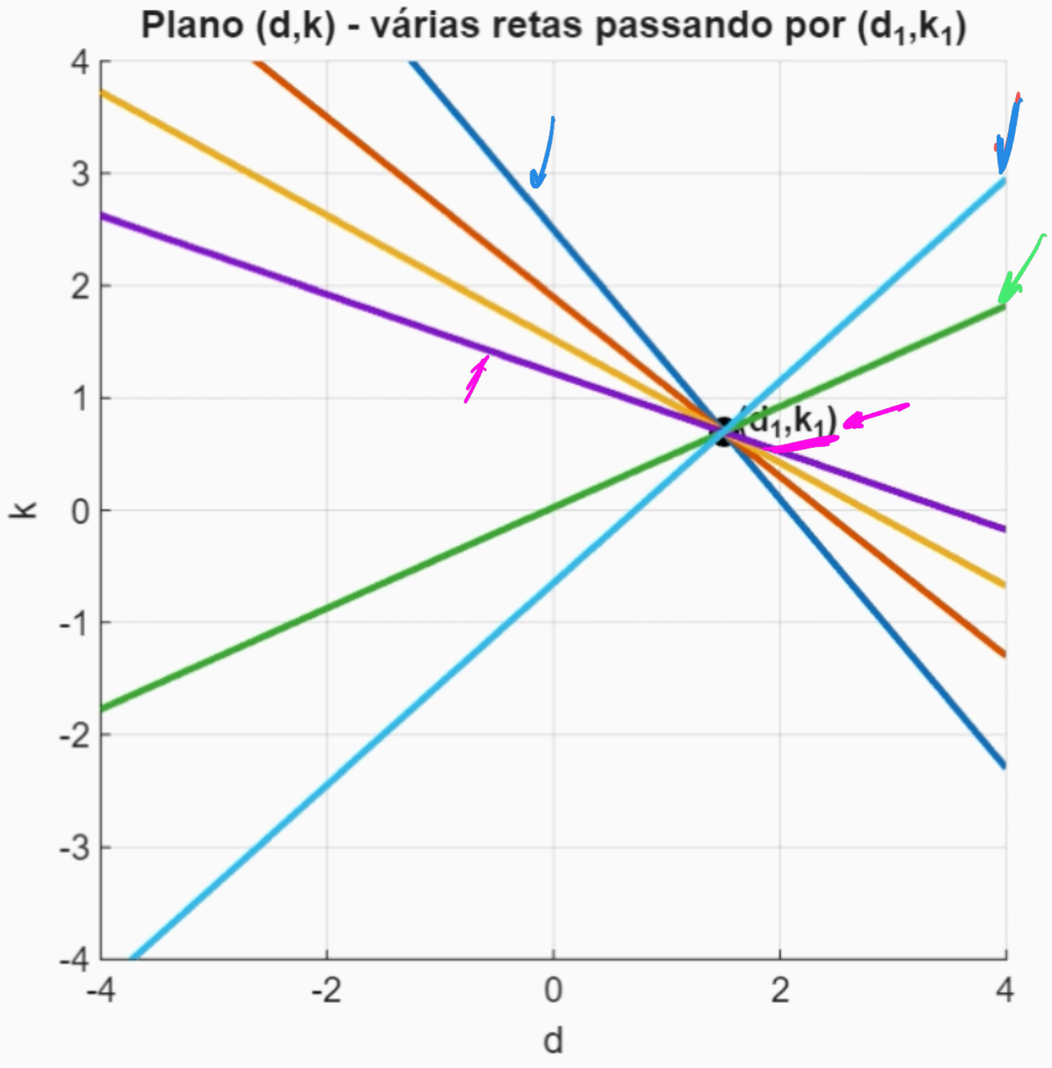
Hough Transform Parameters Space

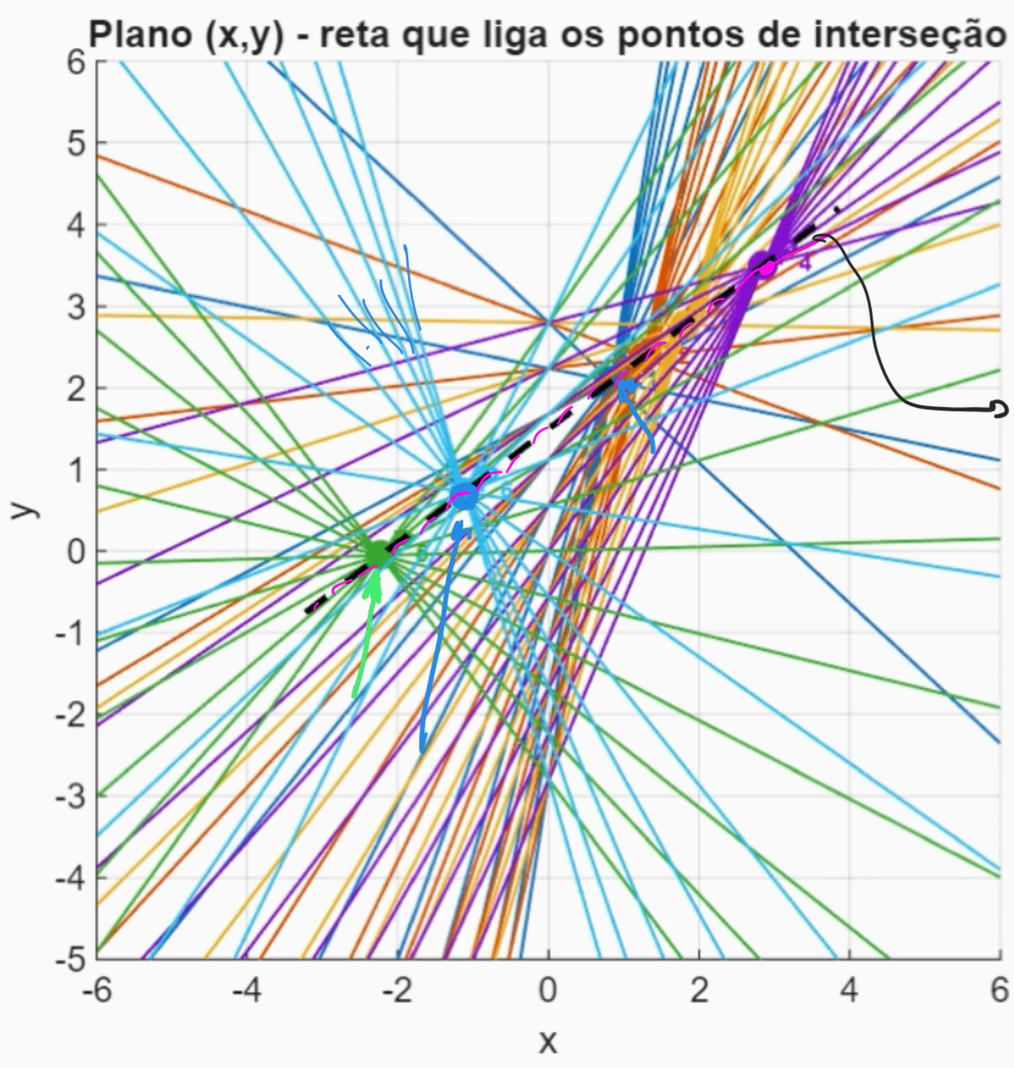


xy space domain


Plano (d,k) - reta selecionada 4
(d1,k1)
k
d
4
3
2
1
0
-1
-2
-3
-4
-4
-2
0
2
4


Plano (x,y) - família colorida 4
y
x
6
5
4
3
2
1
0
-1
-2
-3
-4
-5
-6
-4
-2
0
2
4
6





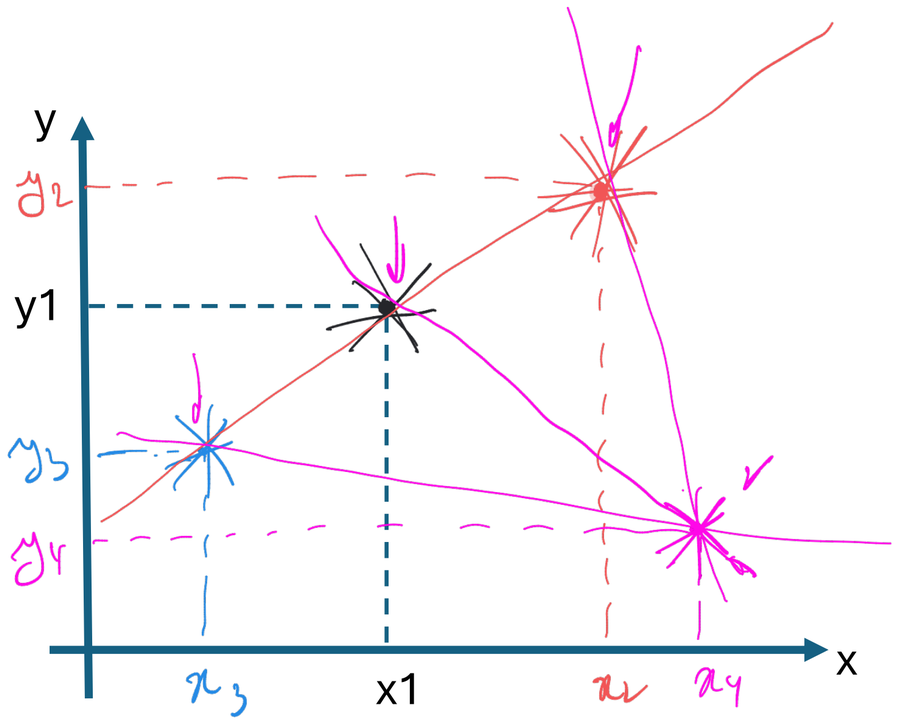
$y = k_1 x + d_1$

parameters space

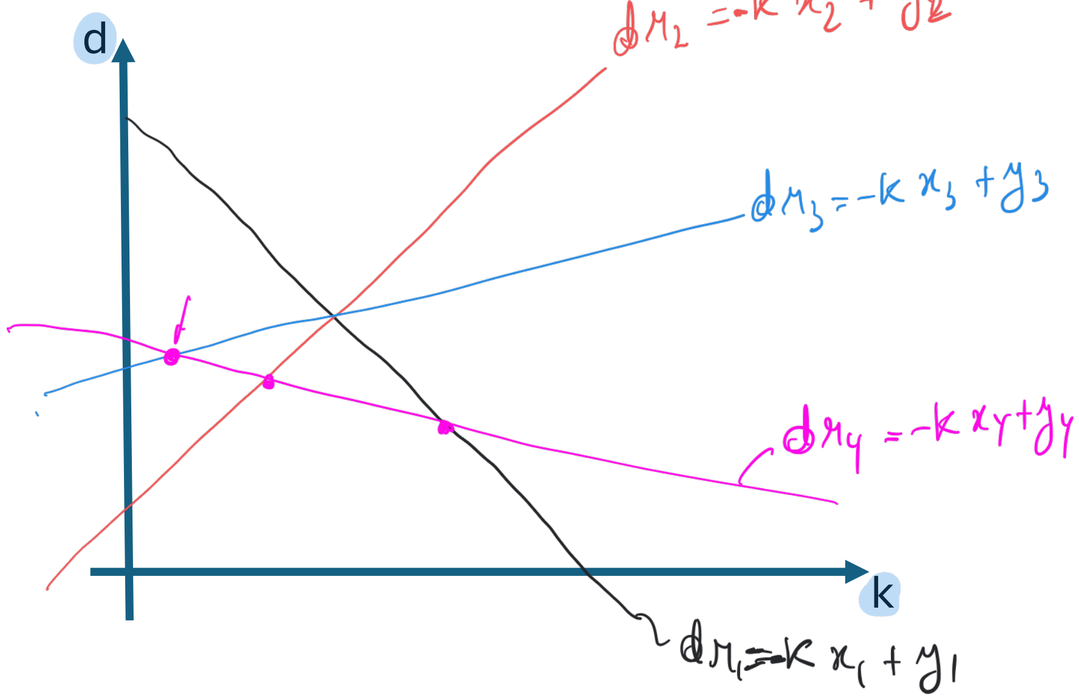
plano xy

Image enhancement in the space domain

Hough Transform



xy space domain



Hough Transform Parameters Space

Figura 1: o ponto roxo não está alinhado

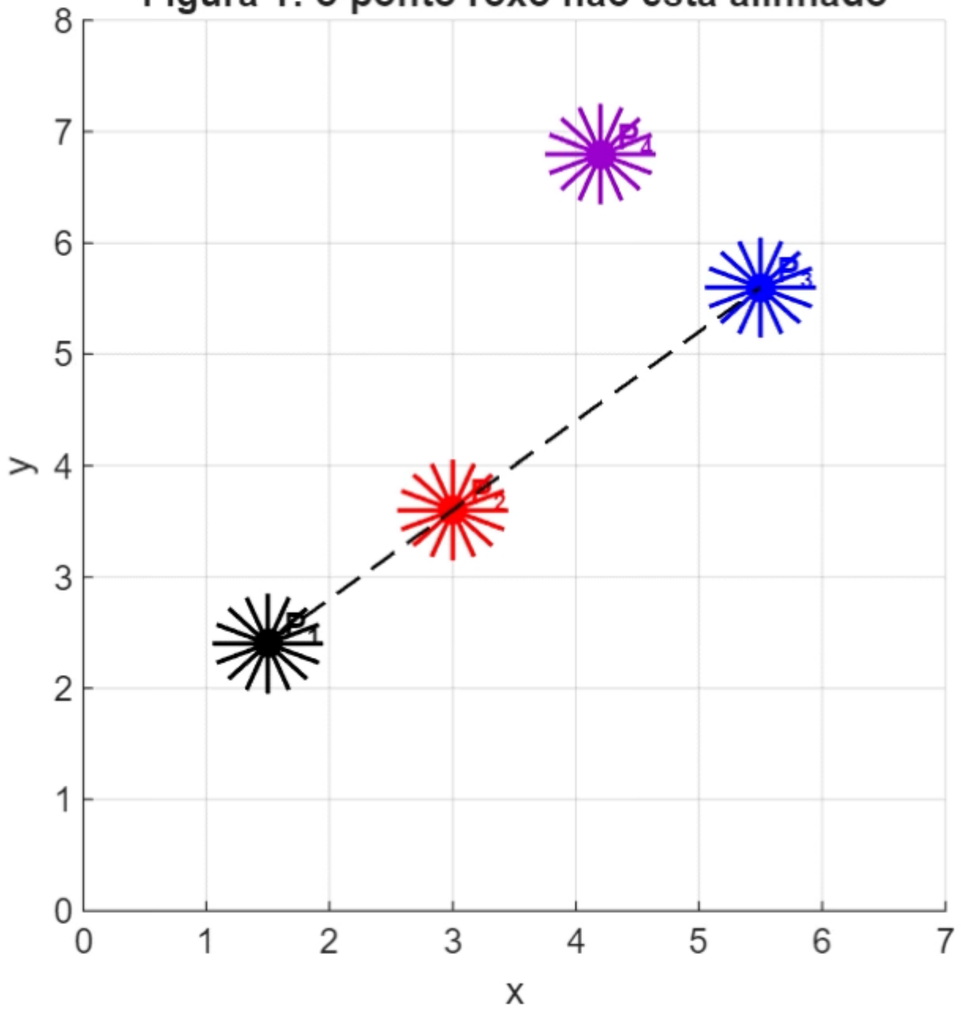


Figura 2: cruzamentos da reta roxa com as outras

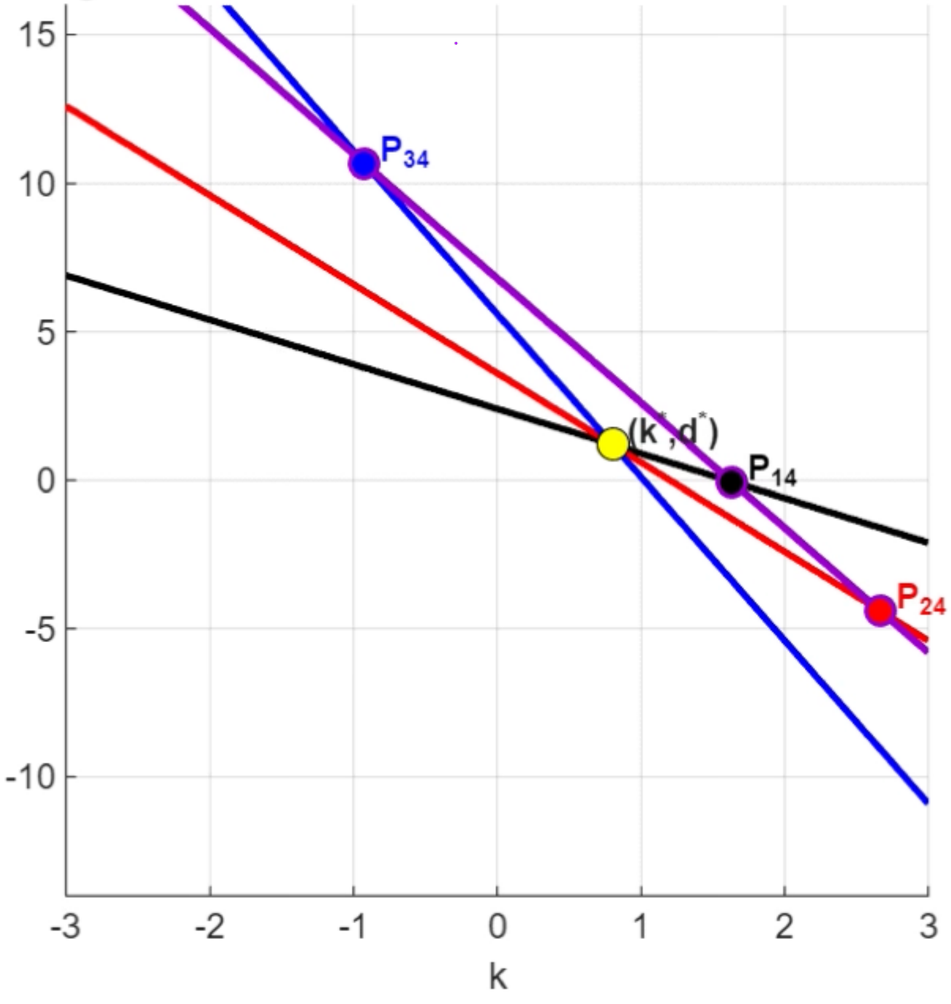
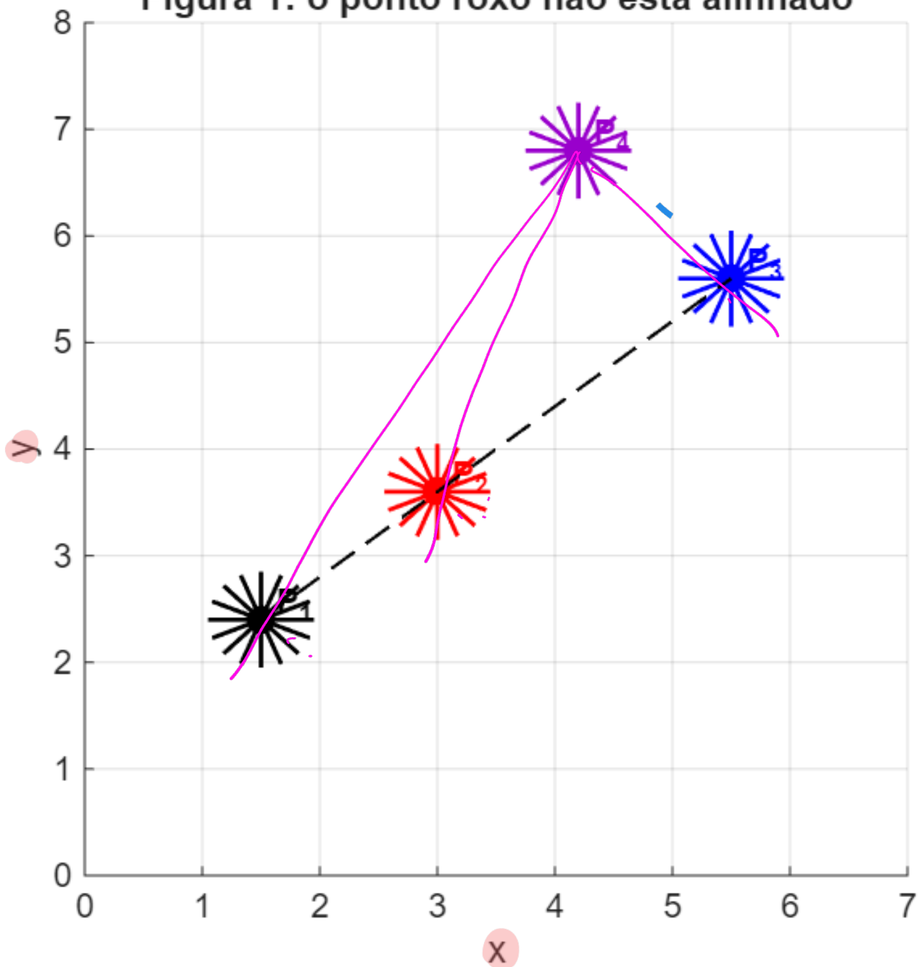


Image enhancement in the space domain

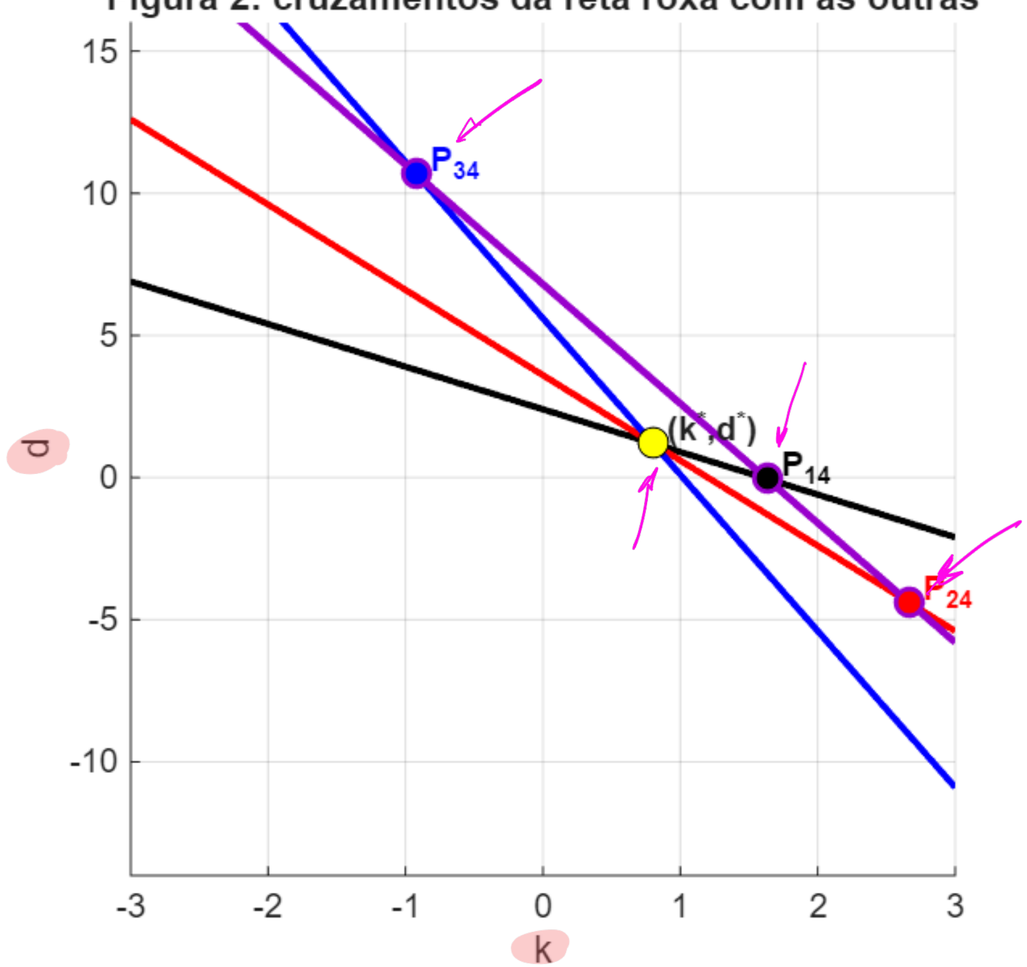
Hough Transform

Figura 1: o ponto roxo não está alinhado



x y domain

Figura 2: cruzamentos da reta roxa com as outras

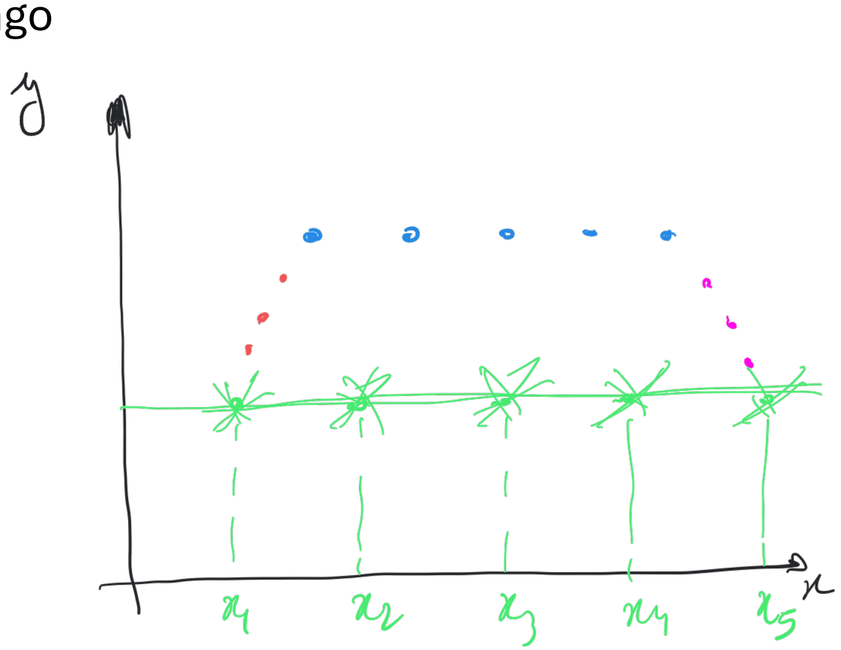


parameter domain

Image enhancement in the space domain

Hough Transform

Exemplo losango



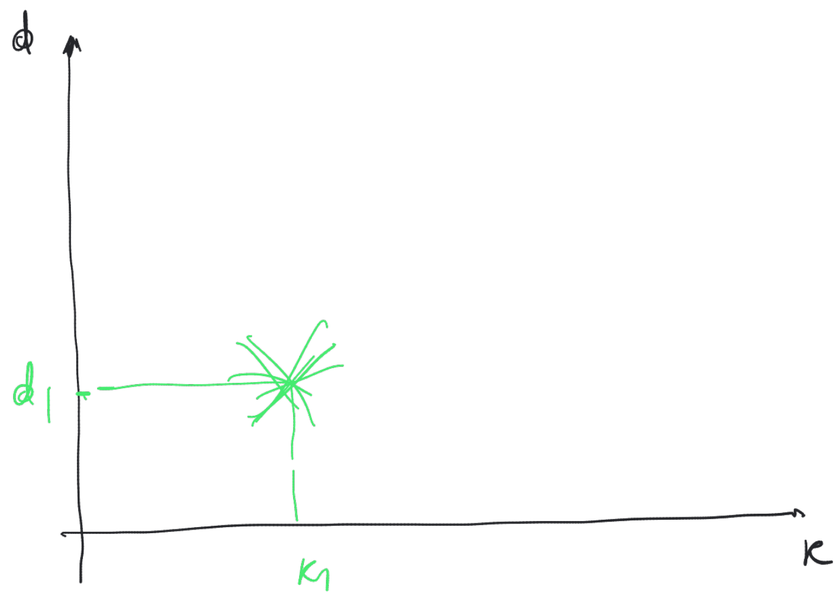


Figura 1: lados do trapézio

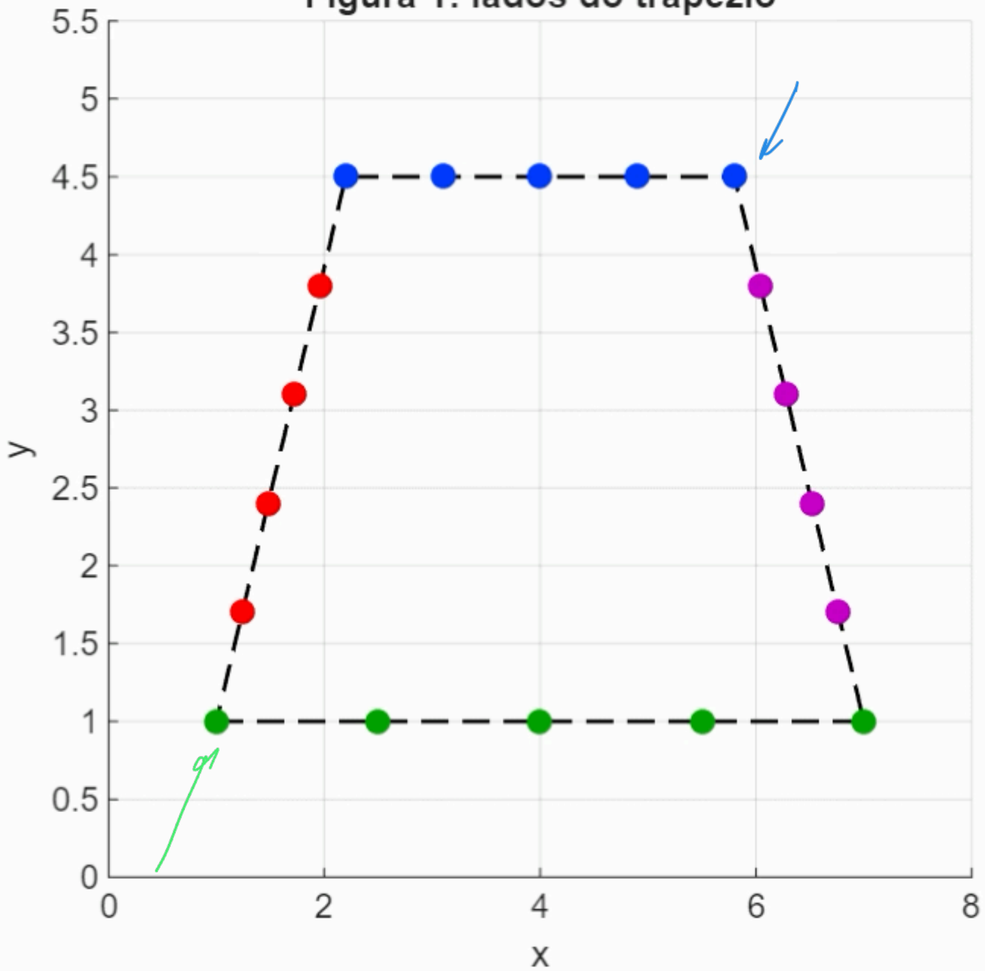
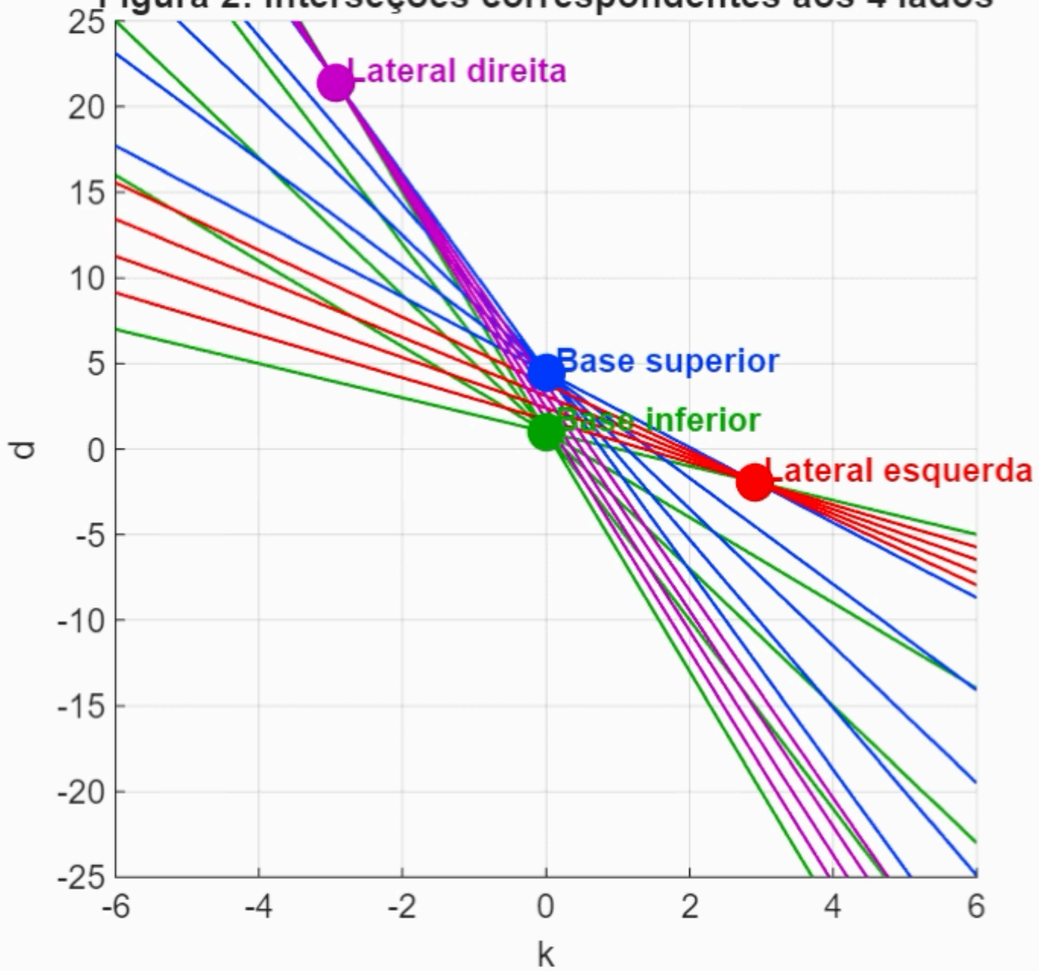
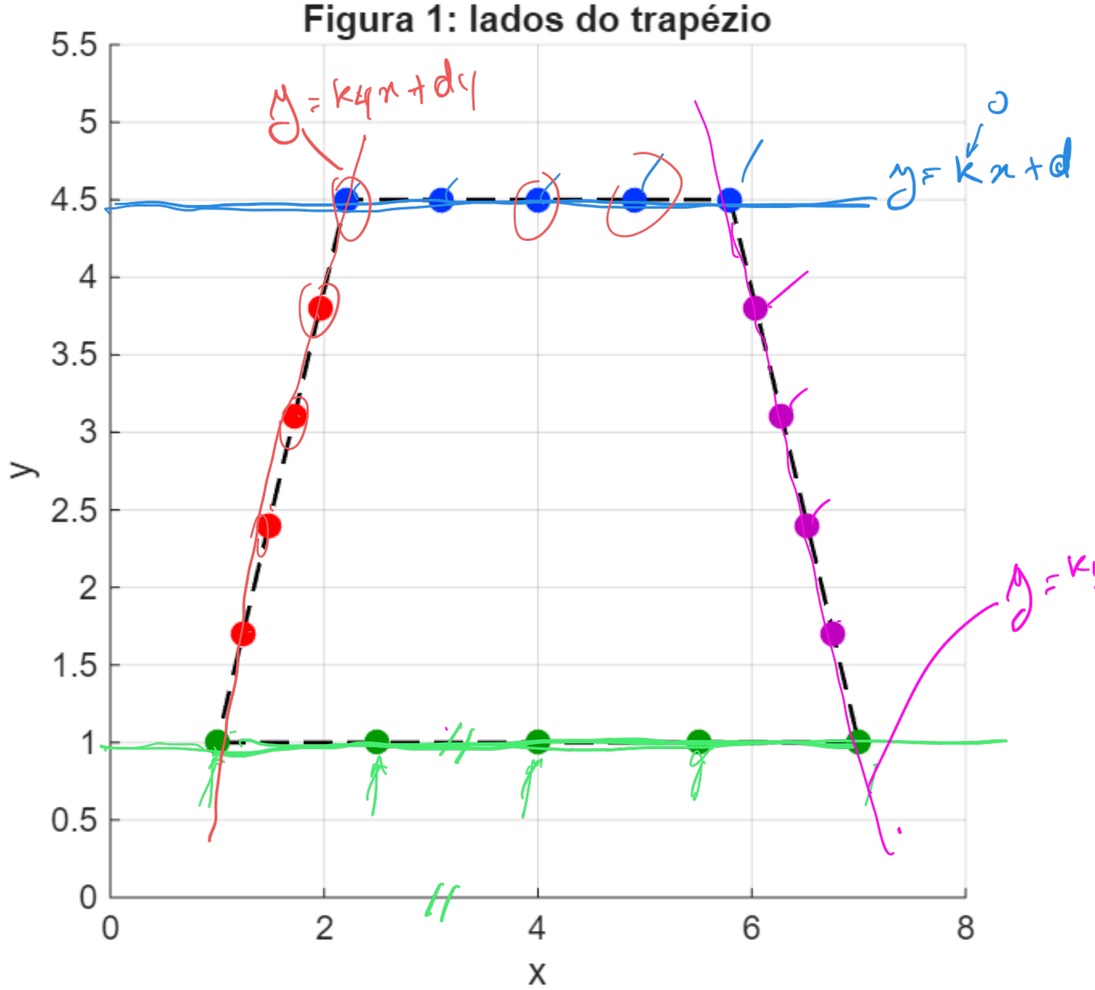
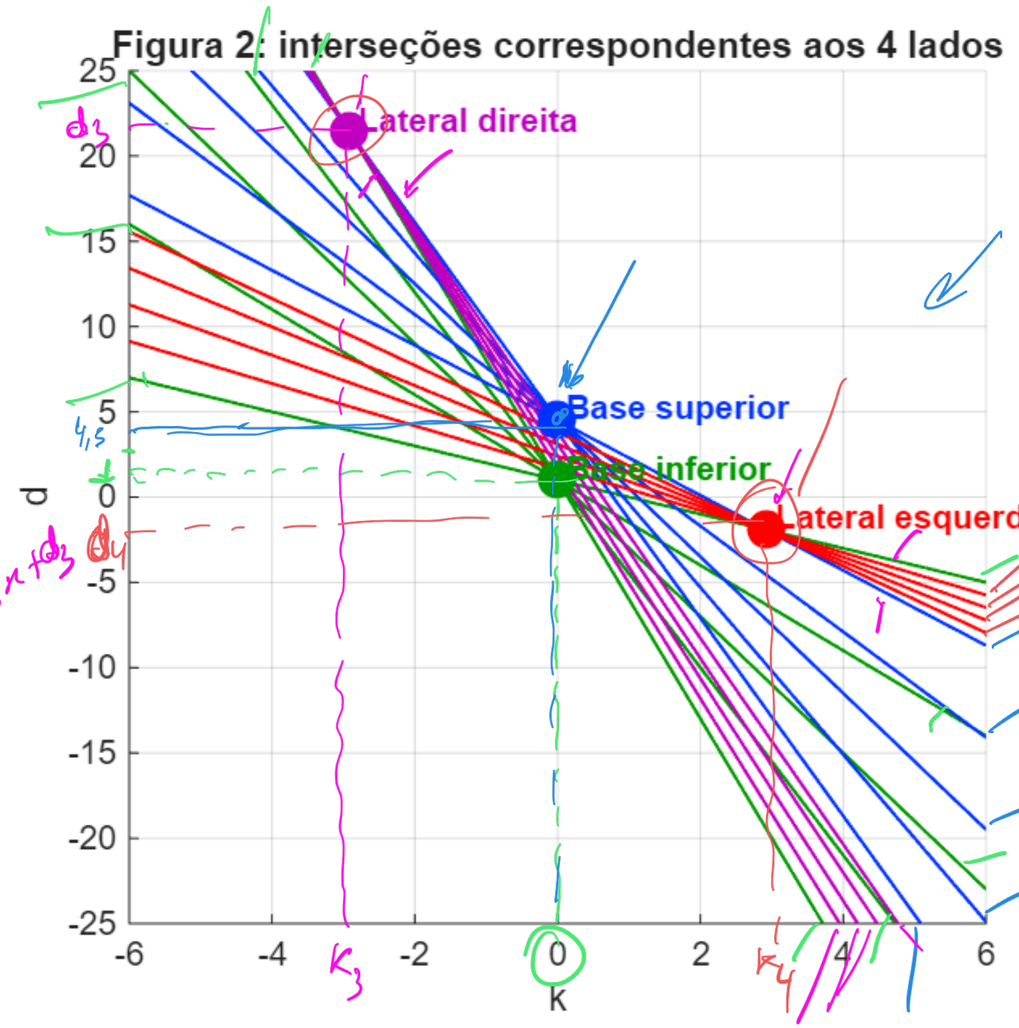


Figura 2: interseções correspondentes aos 4 lados





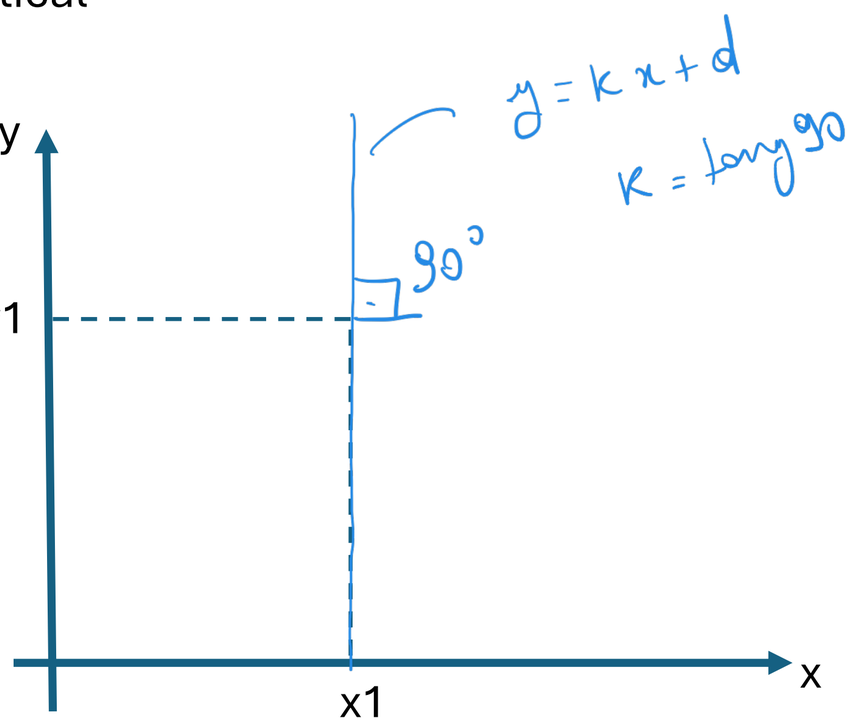


TH_Trapezio_GIF.m

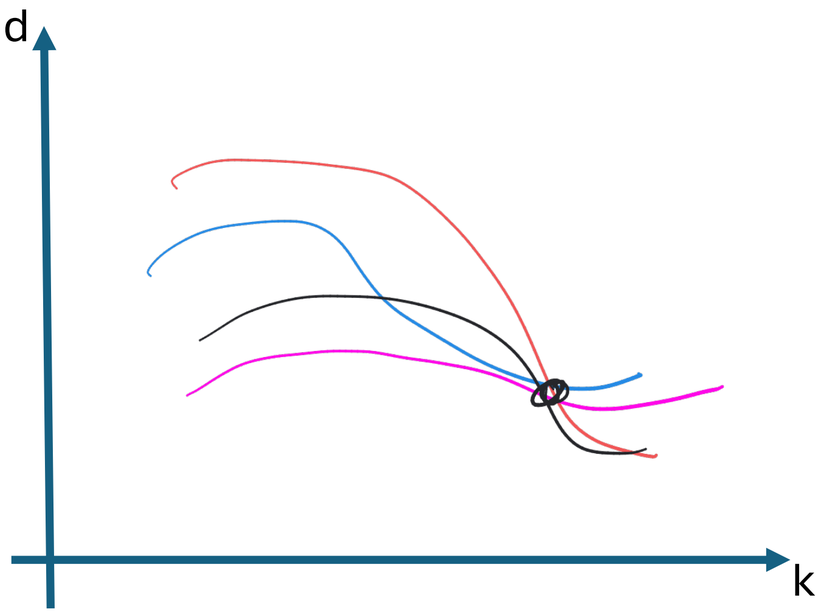
Image enhancement in the space domain

Hough Transform

Reta vertical



xy space domain

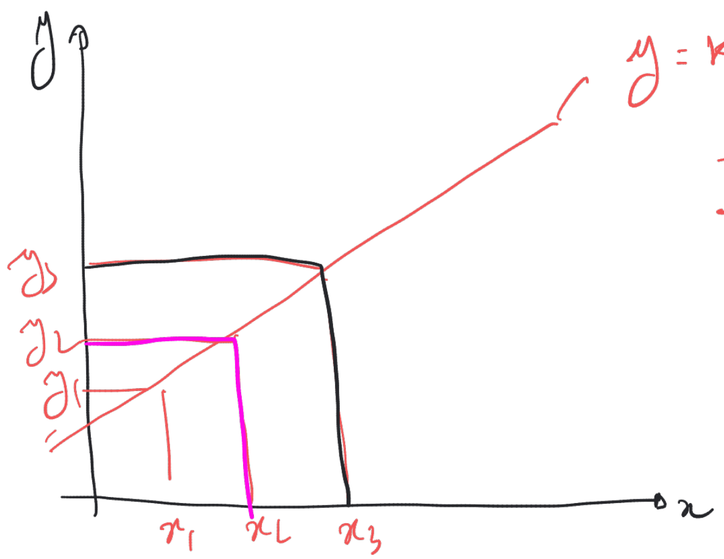


Hough Transform Parameters Space

Image enhancement in the space domain

Hough Transform

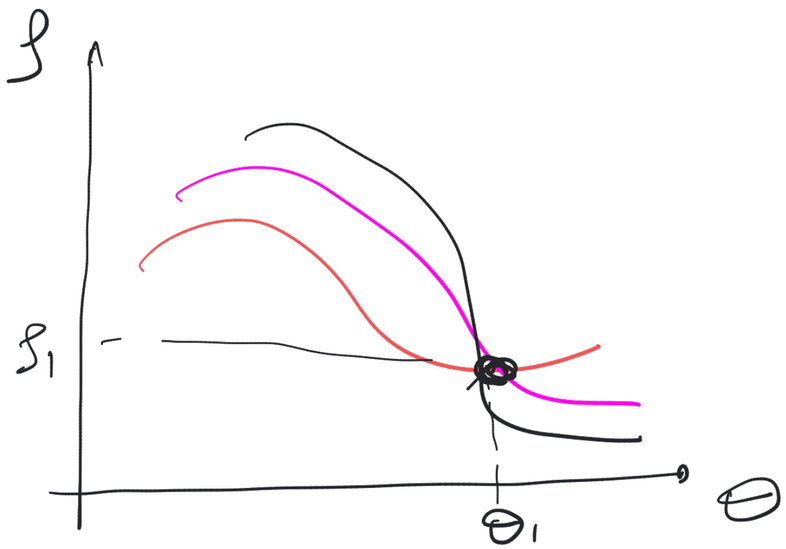
Representação de uma reta em coordenadas polares

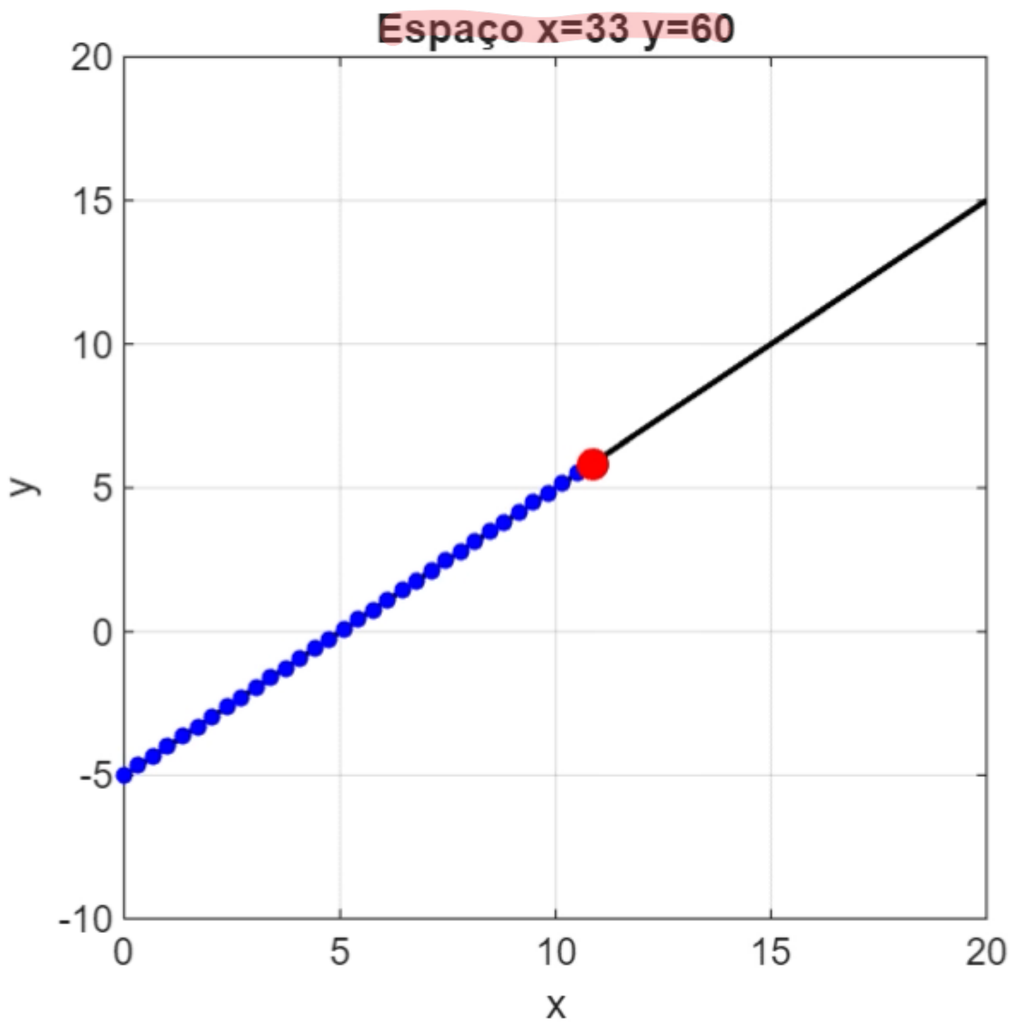


$y = kx + d$

$\rho = x\cos\theta + y\sin\theta$

cada ponto $P(x,y)$ vira uma senoide no plano espaço $\rho \times \theta$




Espaço x=33 y=60
y
x
20
15
10
5
0
-5
-10
0
5
10
15
20

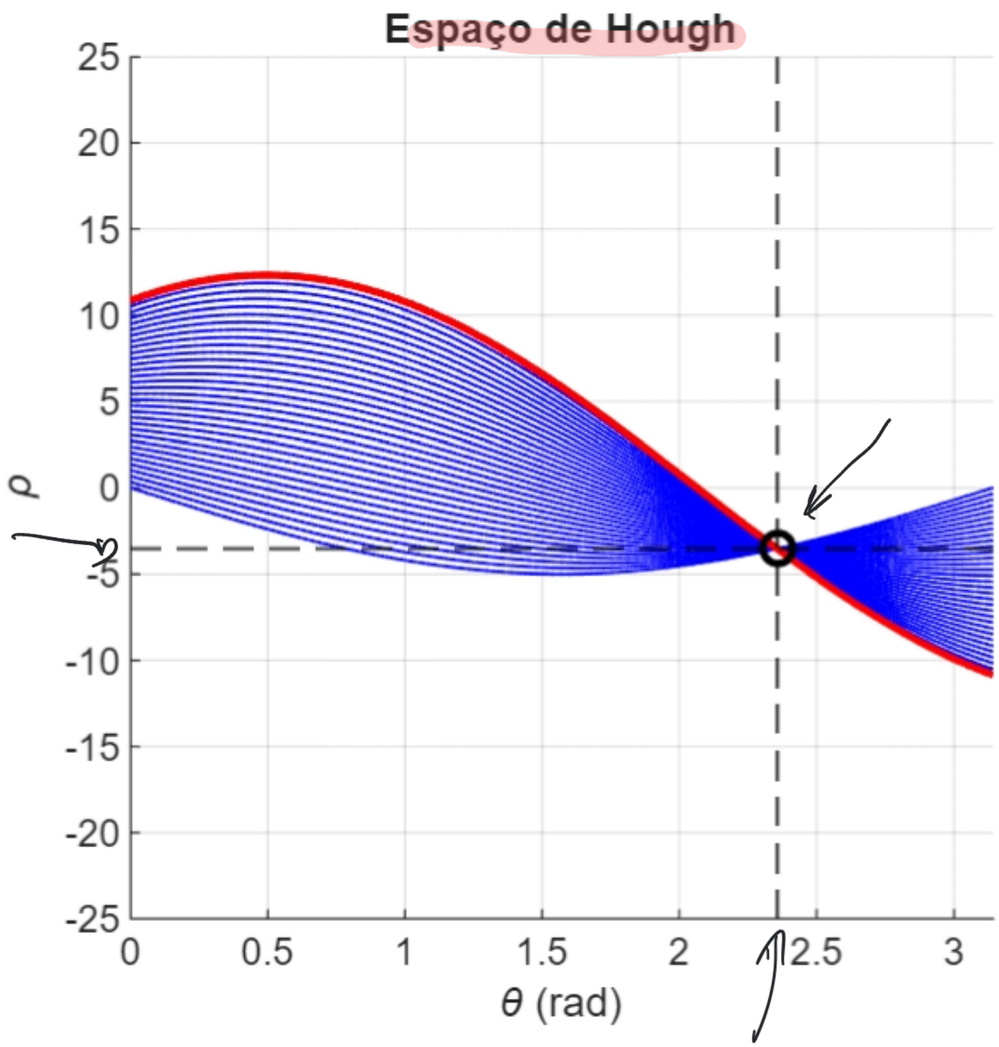
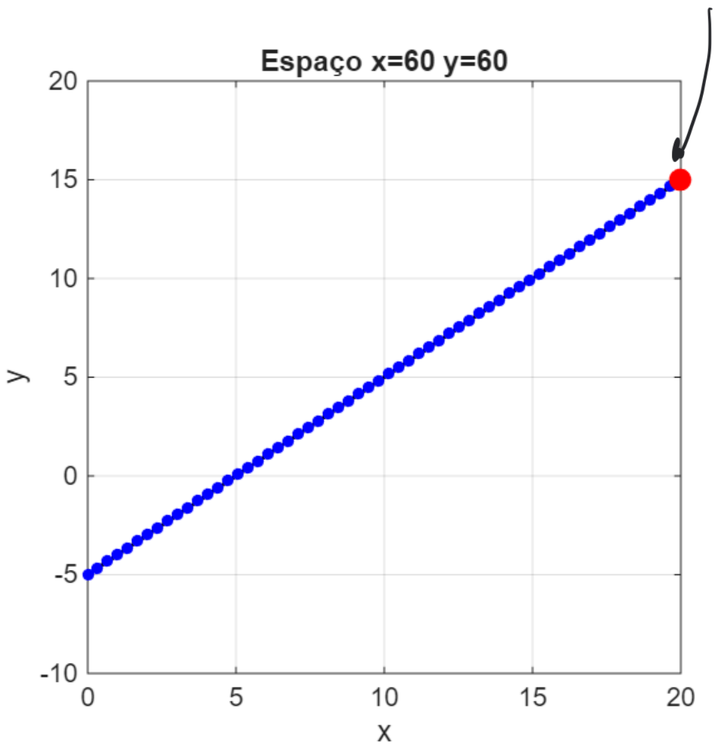

Espaço de Hough
ρ
θ (rad)
25
20
15
10
5
0
-5
-10
-15
-20
-25
0
0.5
1
1.5
2
2.5
3

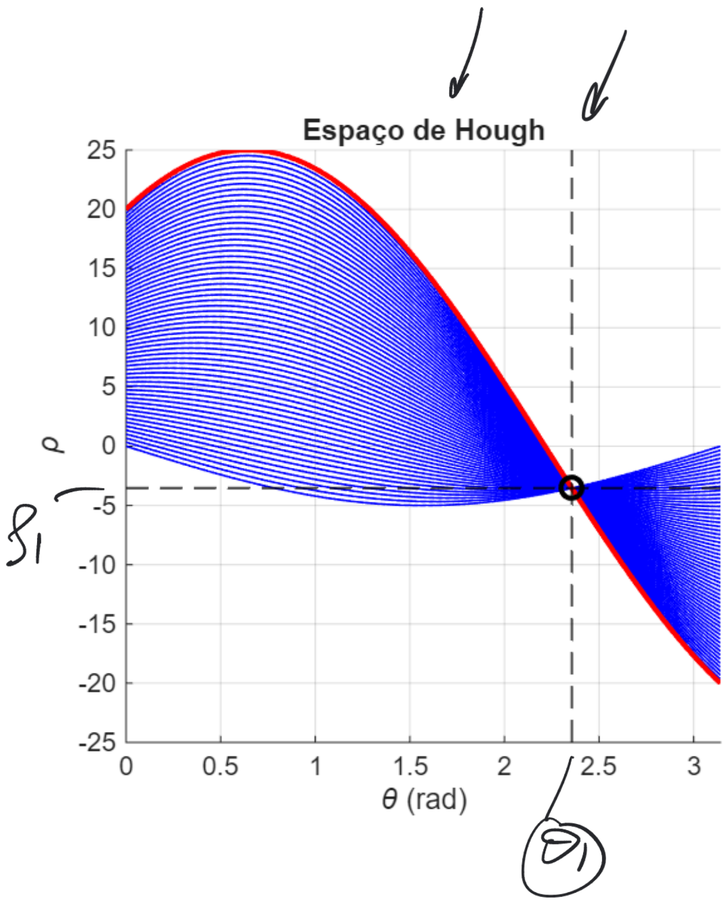
Image enhancement in the space domain

Hough Transform

$$\rho = x\cos\theta + y\sin\theta$$

$$y = -\frac{\cos\theta}{\sin\theta}x + \frac{\rho}{\sin\theta}$$





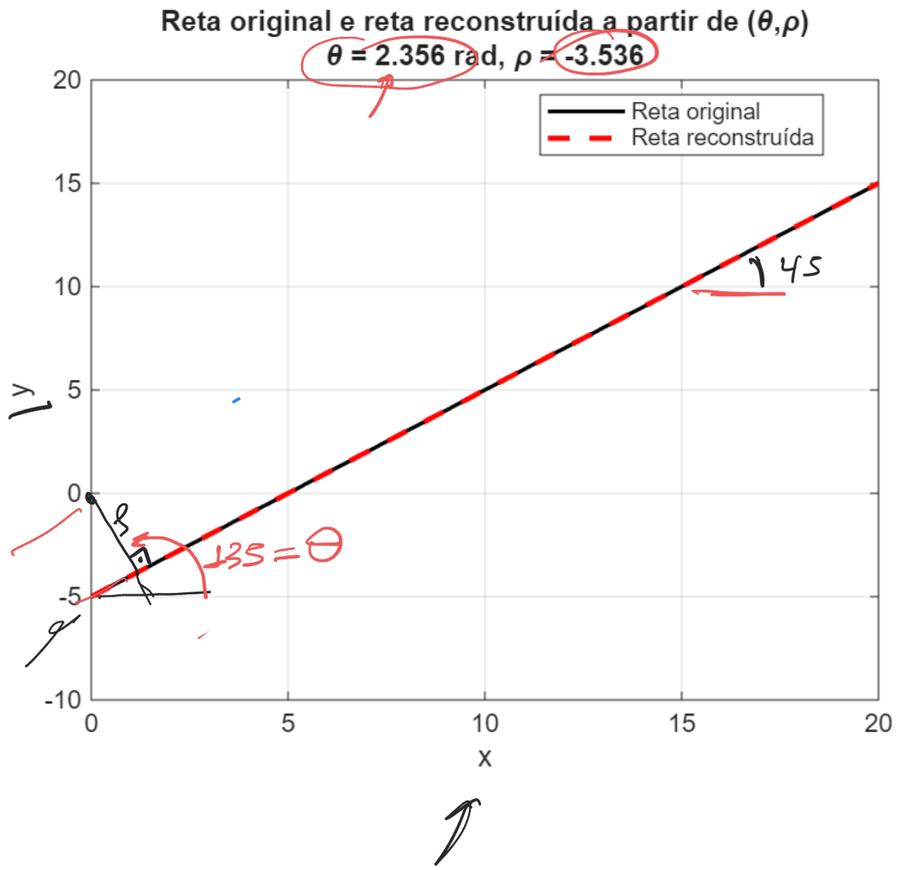


Image enhancement in the space domain

Hough Transform

Similação de deteção de reta ou borda em uma imagem binarizada.

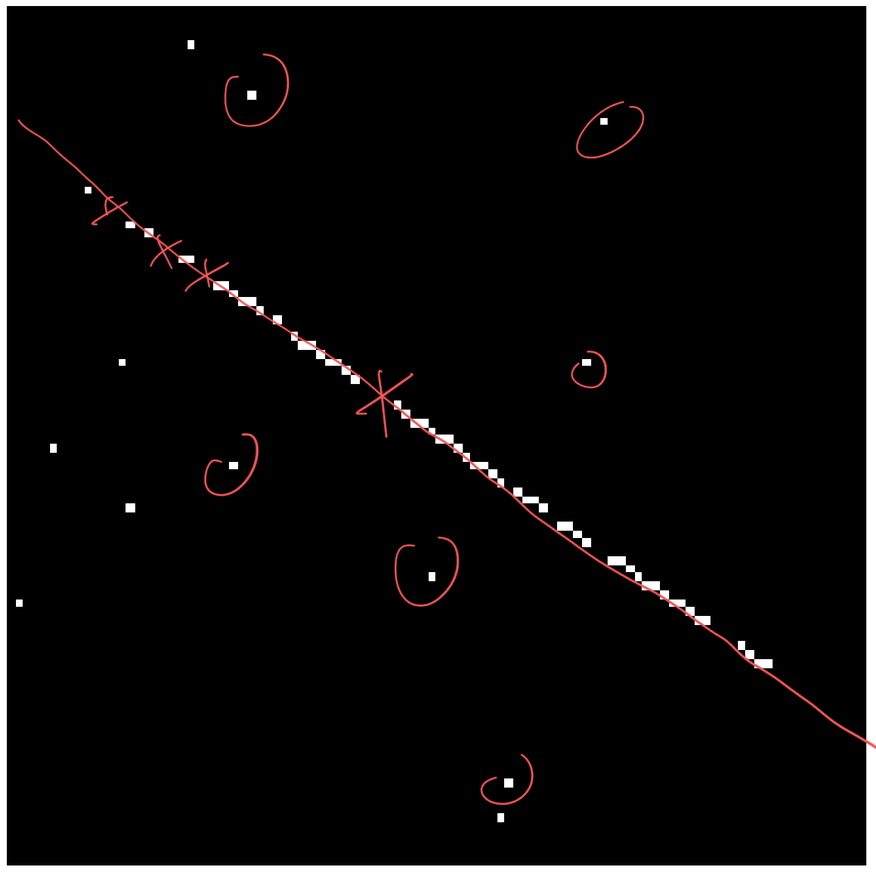



Imagem binária de entrada

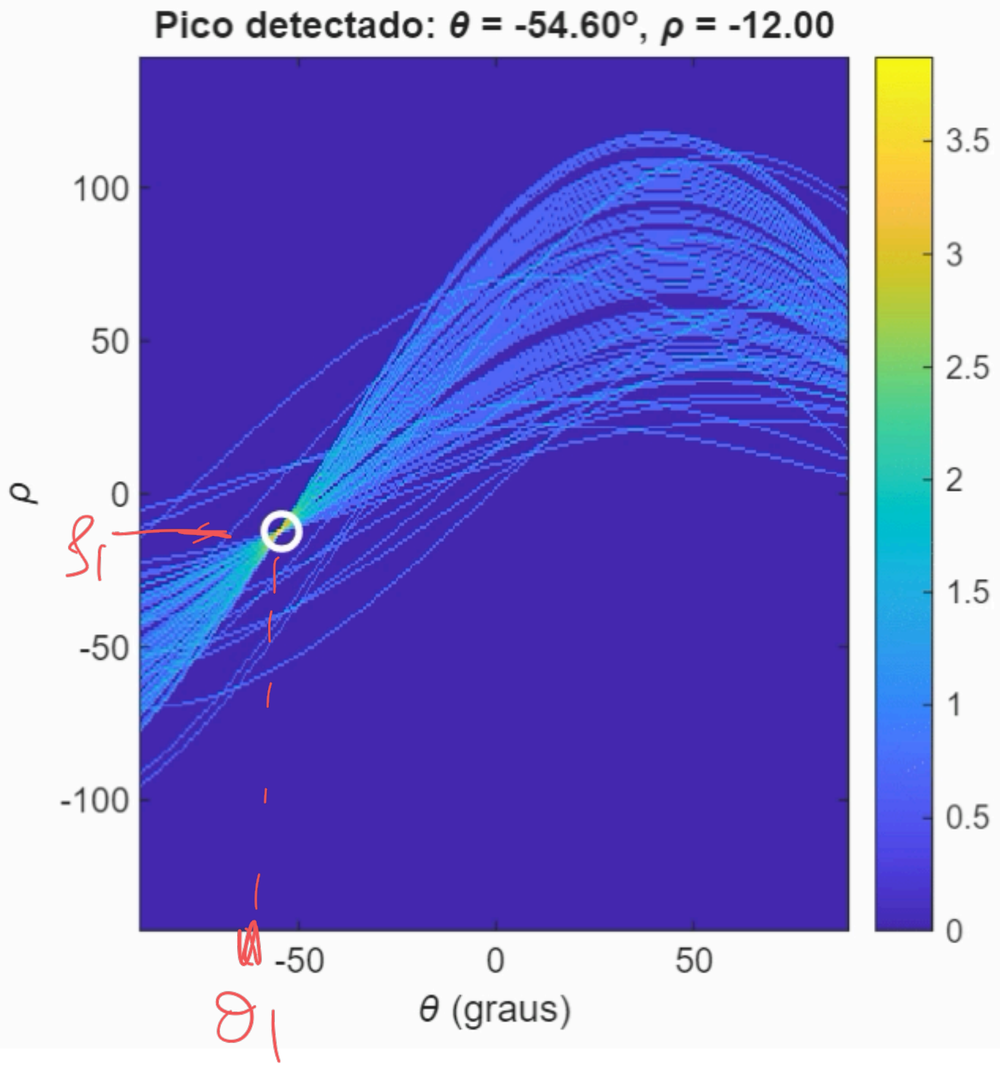

Pico detectado: θ = -54.60°, ρ = -12.00
100
50
0
-50
-100
ρ
-50
0
50
θ (graus)
3.5
3
2.5
2
1.5
1
0.5
0


Imagem binária de entrada

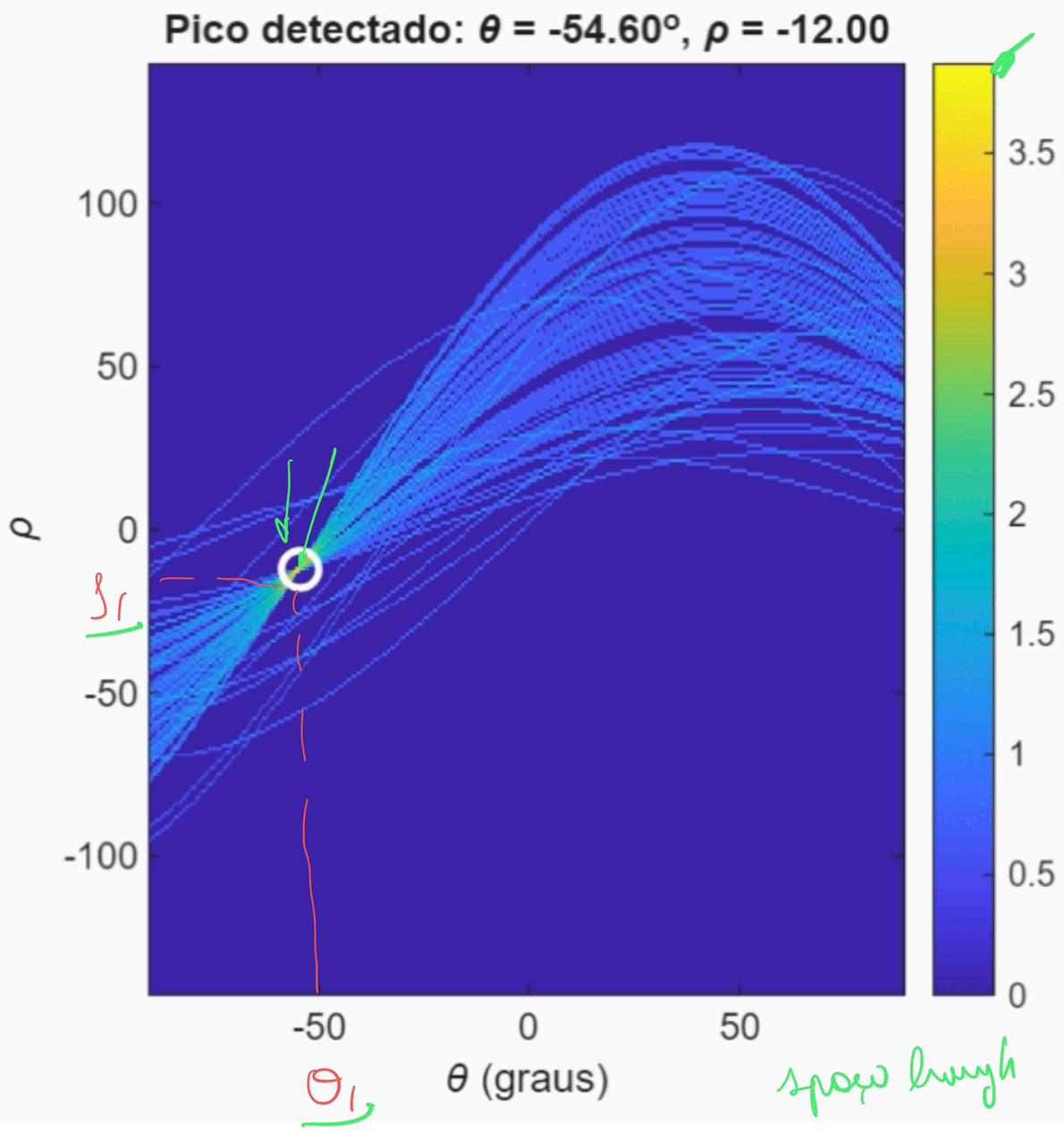

Pico detectado: θ = -54.60°, ρ = -12.00
100
50
0
-50
-100
ρ
-50
0
50
θ (graus)
3.5
3
2.5
2
1.5
1
0.5
0

Image enhancement in the space domain: Hough Transform

MATLAB function for Hough Transform: hough(BW)

```
BW = logical(BW);
```

A imagem de entrada para o comando hough() deve ser B&W.

```
[H, theta, rho] = hough(BW);
```

H: acumulador de votos
theta: ângulas testados
rho: distâncias testadas

H é um acumulador. No espaço de Hough (ρ x θ), o valor de H representa o número de curvas senoidais que intersectam o ponto $P(\rho, \theta)$, curvas do tipo $\rho = x\cos(\theta) + y\sin(\theta)$.

Cada pixel branco na imagem gera uma curva no espaço de Hough (espaço de parâmetros) e contribui com um voto para cada ponto ao longo dessa curva.

Assim se duas senoides passam por $P(\rho, \theta)$, então H=2.
Se 15 senoides passam por esse mesmo ponto, então H=15.

Pontos com valores altos em H indicam possíveis retas na imagem

Image enhancement in the space domain: Hough Transform

MATLAB function for Hough Transform: houghpeaks(H,N)

```
peaks = houghpeaks(H, 3);
```

Retorna os 3 picos mais fortes no acumulador H

Image enhancement in the space domain: Hough Transform

MATLAB function for Hough Transform: houghlines(BW, theta, rho, peaks)

```
lines = houghlines(BW, theta, rho, peaks);

lines(k).point1   % ponto inicial [x y]       % k é o índice de cada reta detectada
lines(k).point2   % ponto final [x y]
lines(k).theta    % ângulo da reta
lines(k).rho      % distância da reta
```

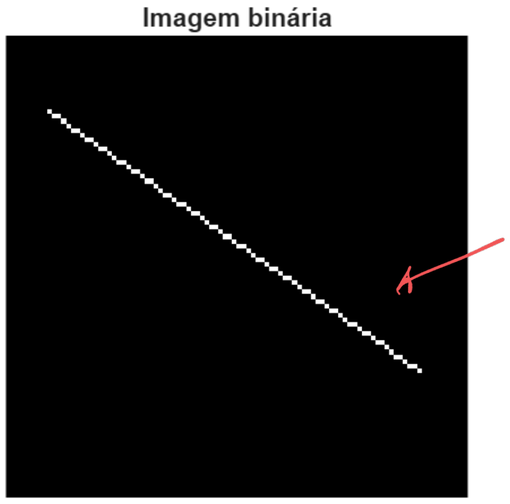
Convete os picos no espaço de Hough (ρ x θ) em segmentos de reta na imagem, ou seja, sai do espaço Hough e volta para a imagem.

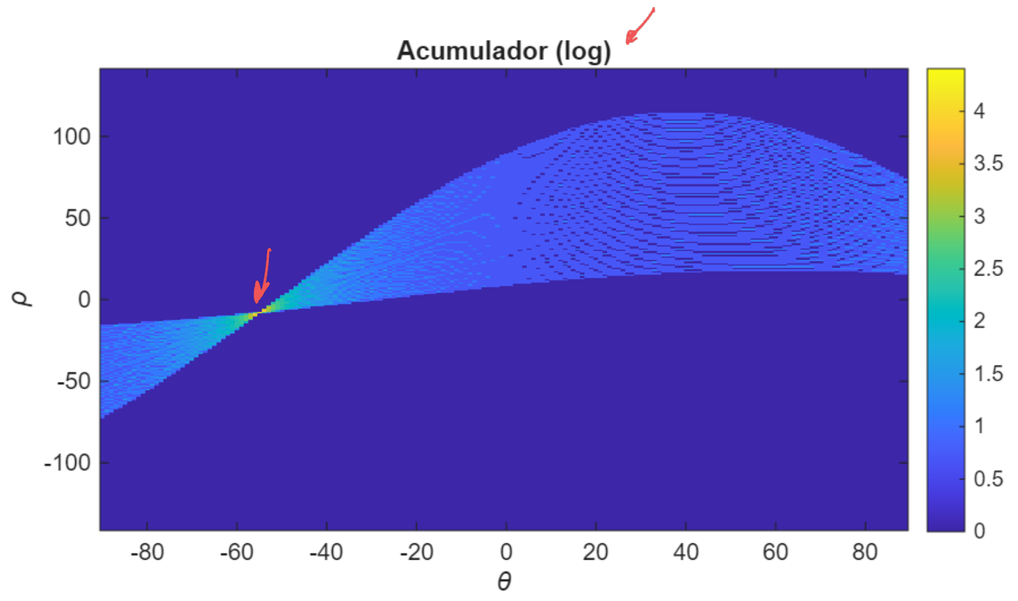
BW: imagem binária (bordas)
theta: vetor de ângulos
rho: vetor de distâncias
peaks: índices dos picos no acumulador

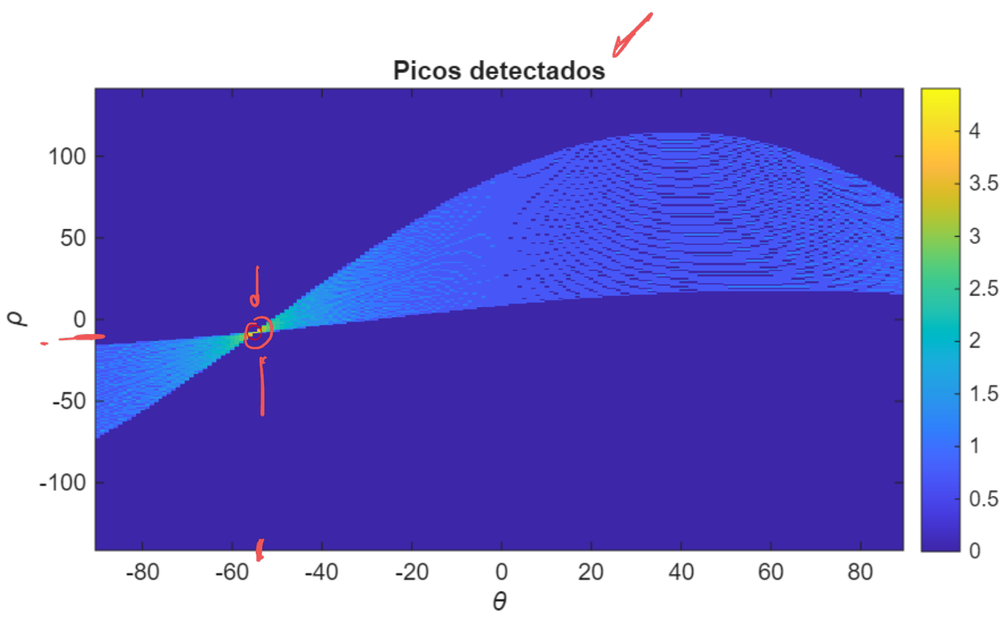
A função pega cada pico, encontra os pixels alinhados e monta um segmento de reta.

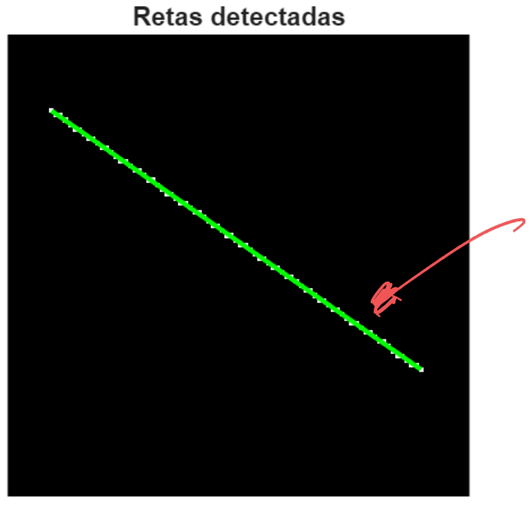
```
xy = [lines(k).point1; lines(k).point2];
plot(xy(:,1), xy(:,2), 'g-', 'LineWidth', 2);
```

Image enhancement in the space domain: Hough Transform





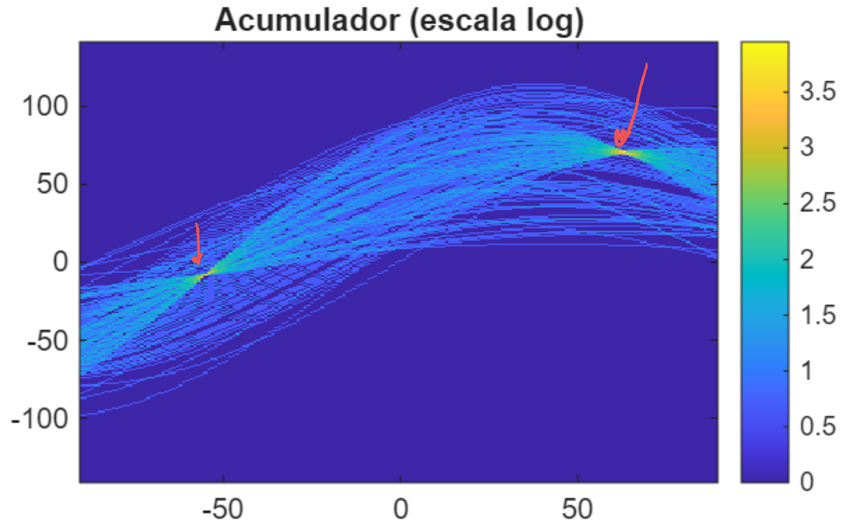


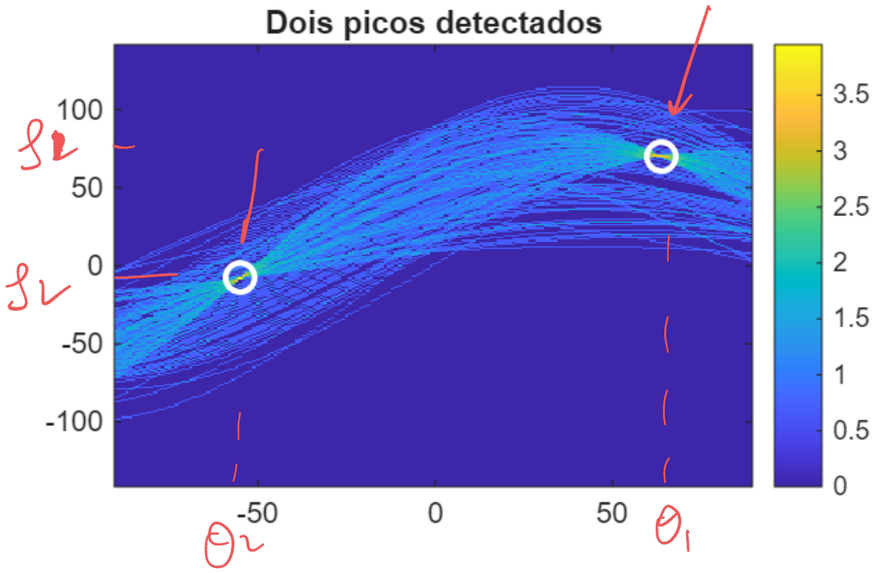


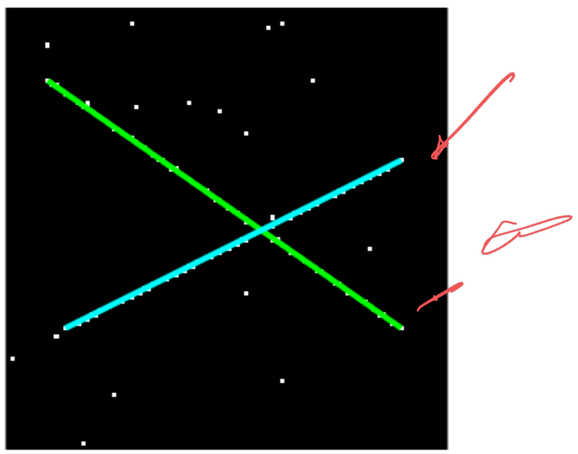
HT_Using_Matlab_Functions.m

Image enhancement in the space domain: Hough Transform



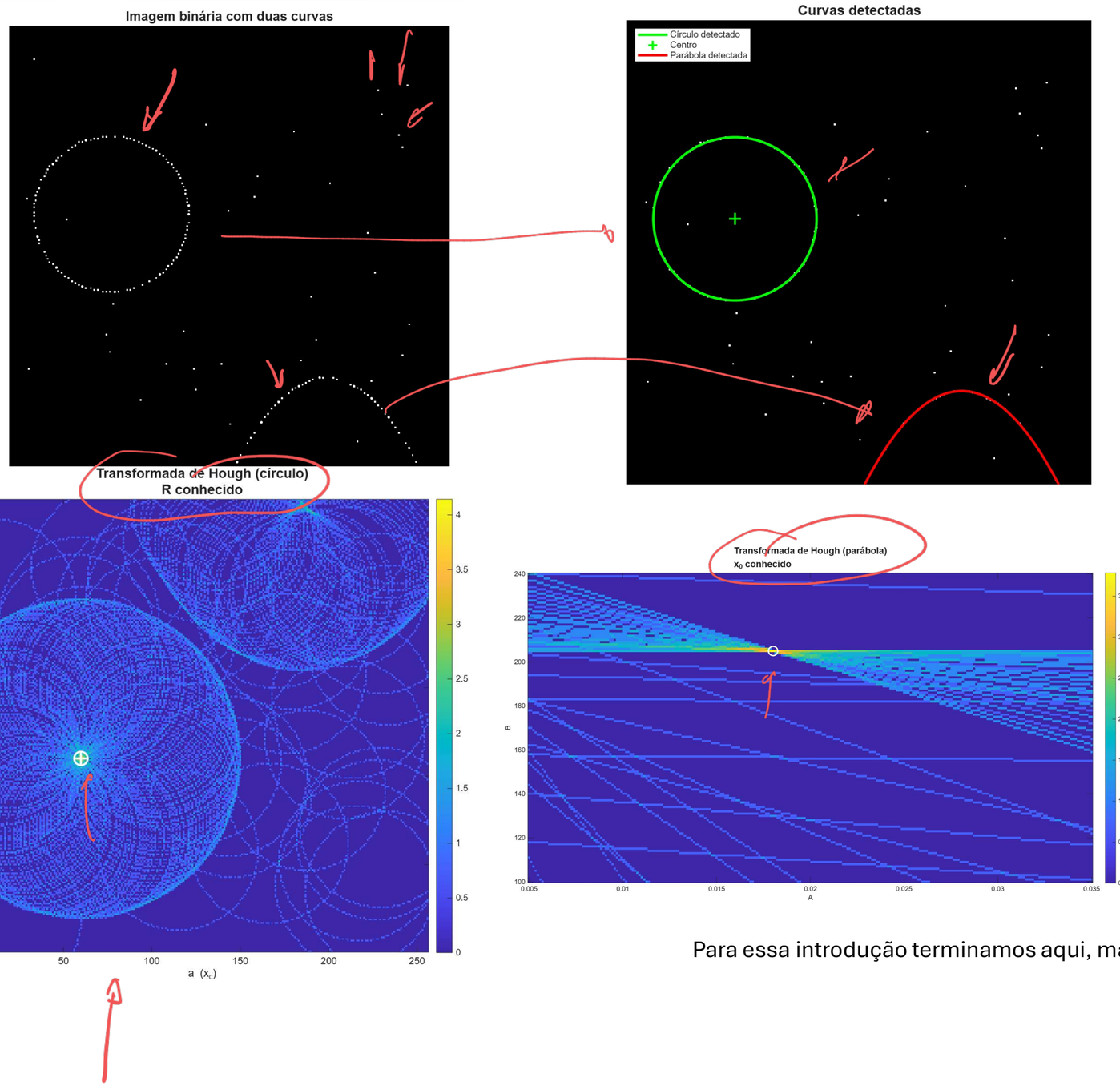






HT_Using_Matlab_Functions_Duas_Retas.m

Image enhancement in the space domain: Hough Transform



Para uma reta são dois parâmetros:

(ρ, θ)

Para um círculo são necessários de três parâmetros:

$(x-a)^2 + (x-b)^2 = R^2$

$H(a, b, R)$

Fixando R, podemos visualizar uma seção do espaço de Hough no plano (a, b).

Parábola requer três parâmetros: A, x_0 *e* B:

$y = A\,(x - x_0)^2 + \mathrm{B}$

$H(A, x_0, B)$

Na figura fixamos x_0: $H(A, B \mid x_0)$

Na medida que aumentamos o número de parâmetros necessários para descrever uma forma, aumenta a dimensionalidade do espaço de parâmetros da HT também aumenta.

Para essa introdução terminamos aqui, mas existe ainda a **Generalized Hough Transform**